

Chapter 2. Networking and Internetworking

DISTRIBUTED SYSTEMS DEVELOPMENT

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Distributed Systems: Concepts and Design
Edition 5, ©Addison-Wesley, 2012



Distributed Systems (I)

- Distributed systems use local area networks, wide area networks, and interworks for communication.
- The performance, reliability, scalability, mobility, and quality of service characteristics of the underlying networks impact the behavior of distributed systems and hence their design.
- User demands resulted in the emergence of wireless network and QoS guarantees.
- Solutions:
 - protocol layering, packet switching, routing and data streaming.
 - Integration of heterogeneous systems
 - IPV4 to IPV6 to address enormous growth in space
 - Improvements in mobility, security, and QoS
- Standards: Ethernet, IEEE 802.11 (WiFi), Bluetooth

Distributed Systems (2)

- Networks in a distributed system are built from a variety of transmission media: wire, cable, fiber, wireless channels; hardware devices bridges, routers, hubs, repeaters, etc.
- Software including protocol stacks, communication handlers and drivers;
- These form the communication subsystem.
- Computers on the communication subsystem are known as hosts;

Distributed Systems (3)

- The Internet is a single communication subsystem providing communication between hosts
- The Internet is constructed of many subnets
- Chapter 3 provides an overview of communication requirements of distributed systems

Networking Issues in Distributed Systems

- It began with a simple applications and their requirements: file transfer, email, newsgroup.
- It has moved to hotly debated social applications with not only stringent security requirements but also “fact or lies” type of requirement.
- Lets look at some common requirements:

Performance

- Latency and point-to-point transfer rate.
- Latency: the delay that occurs after a send operation is executed and before data starts to arrive at the destination computer.
- Data Transfer rate: the speed at which data can be transferred between two computers once transmission has begun, specified in bits per second.
- Combining:

Message tx time = latency + length/data transfer rate

- Total system bandwidth of a network is a measure of throughput– the total volume of traffic that can be transferred across the network in a given time.
- For example, in Ethernet full transmission capacity is used for every transmission capacity of the network.
- Therefore in Ethernet, bandwidth is same a data transfer rate.
- In other networks with different channels sending simultaneously, total bandwidth and transfer rates are different.

Scalability

- Computer networks are an indispensable part of infrastructure of modern societies.
- Industry experts now predict that the number of Internet-connected devices will exceed 15 billion nodes by 2015 and top 50 billion by 2020. Electronic Weekly, 2013.
- It is now in the order of trillion nodes.
- For IPv4, this pool is 32-bits (2^{32}) in size and contains **4,294,967,296** IPv4 addresses. The IPv6 address space is **128-bits** (2^{128}) in size, containing **340,282,366,920,938,463,463,374,607,431,768,211,456** IPv6 addresses.” -- <https://www.ripe.net/about-us/press-centre/understanding-ip-addressing>
- How do you scale to this many devices? Simple application such as web depend on the number of users and of course, degree of locality.

Reliability

- Communication errors; Guarantee error-free communication
- Server reliability; fail-over or fail-stop
- Dependability = availability + fault-tolerance + reliability

Security

- First level of defense for many organizations is to protect its networks and computers with a firewall.
- A firewall runs on a gateway computer that stands at the network entry point to an organization's intranet.
- A firewall receives and filters all of the messages organization's security policy to allow certain incoming and outgoing message to pass in and out and reject all others.
- Fine-grained form of security can be realized using cryptographic techniques.
- End-to-end encryption, authentication.

Other requirements

- Mobility: wireless network address the mobility requirement
- Quality of service: Response time and ability to meet deadlines; multi-media guarantees for bandwidth, bounded latencies.
- Multicasting: Broadcast has become common mode of communication. Network has to support the simultaneously transmission of messages to several recipients.

Types of network

- Many types of network evolved to address the requirements:
- Personal area network: WPAN wireless PAN: devices carried by users on them, low power and low energy network.
- Local Area Network: LAN: Ethernet connected computers: 10gbs bandwidth
- Wide area network WAN: WAN carry messages at lower speeds between nodes that are often different organizations and separated by large distances.
- Metropolitan area network (MANs): Based on fiber optic cabling for transmission of video, voice, and other data.
- Equivalent wireless versions of the above: WALNs, WMANs, WWANs, intranet.

Figure 3.1

Network performance

	<i>Example</i>	<i>Range</i>	<i>Bandwidth (Mbps)</i>	<i>Latency (ms)</i>
<i>Wired:</i>				
LAN	Ethernet	1–2 kms	10–10,000	1–10
WAN	IP routing	worldwide	0.010–600	100–500
MAN	ATM	2–50 kms	1–600	10
Internetwork	Internet	worldwide	0.5–600	100–500
<i>Wireless:</i>				
		km		
WPAN	Bluetooth (IEEE 802.15.1)	10–30m	0.5–2	5–20
WLAN	WiFi (IEEE 802.11)	0.15–1.5 km	11–108	5–20
WMAN	WiMAX (IEEE 802.16)	5–50 km	1.5–20	5–20
WWAN	3G phone	cell: 1–5	348–14.4	100–500
	4G		100 (Mbit/s)	
	5G		20 Gigabits peak	

Network Principles: Packet Switching

- Packet switching: Basis for all computer networks is packet switching technique.
 - Packets for different destinations. Packets are queued in a buffer and transmitted when the communication link is available.
 - Communication is asynchronous, packets arrived out of order and need to be ordered.
 - Packets take finite time to travel through the network.

Network Principles: Packet Switching

- Packet transmission
 - In most applications requirement is for the transmission of logical units of information or messages.
 - Before transmission, message is divided into packets of binary data with addressing information for identifying the source and destination.
 - Need to use optimal packet size to allow for buffering, and avoid delays in transmission.

Network Principles: Data Streaming

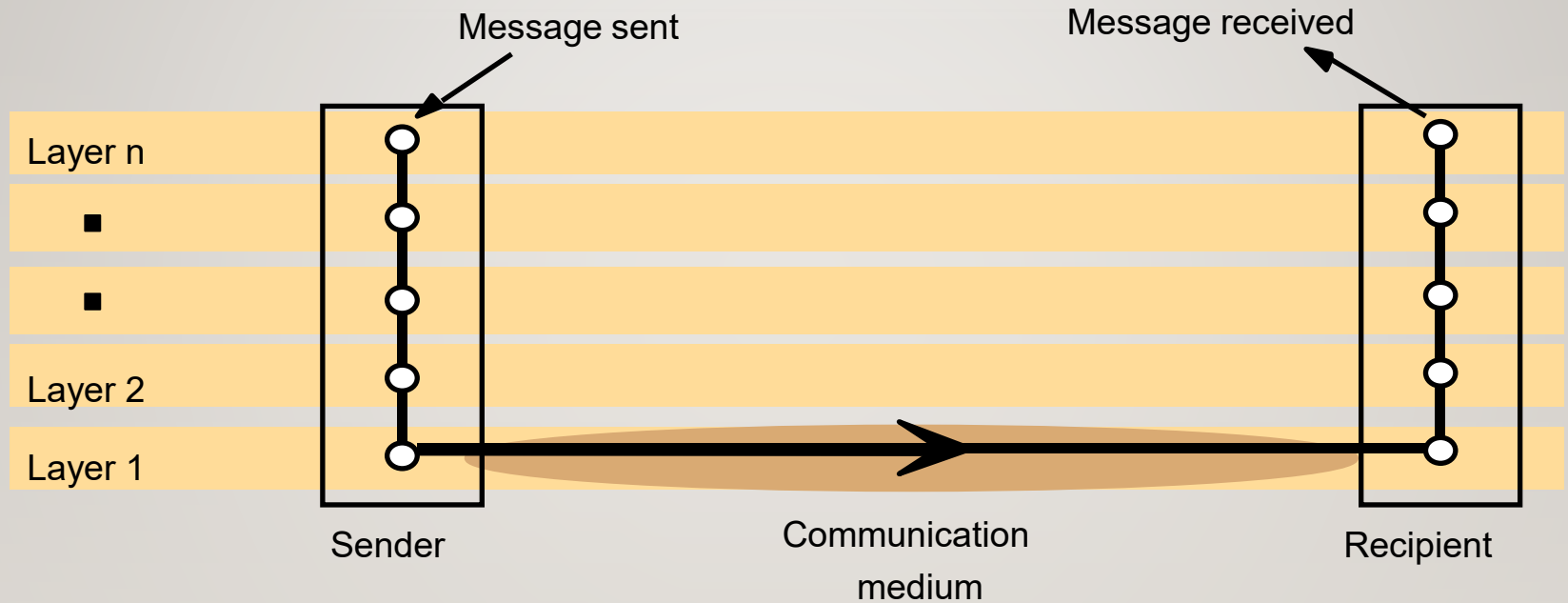
- Transmission for display of audio and video in real time is referred to as streaming.
- Data compression becomes important due to high volumes of data
- Hard deadline for delivery of packets: if it does not arrive in time it is useless
- Data flow needs to be continuous as opposed to intermittent traffic generated by typical client/server system.

Network Principles: Data Streaming

- Timely delivery of audio and video streams depends upon the availability of connections with adequate QoS – bandwidth, latency, and reliability
- Need to reserve channels, and have adequate buffering
- All these are done by upper level protocols at the network and transport layers.
- Lesson: When things get complex create a new layer of abstraction.

Figure 3.2

Conceptual layering of protocol software



Network Principles: Switching Schemes

- A network consists of a set of nodes connected together by circuits. To transmit information between two arbitrary nodes, a switching system is required.
- Broadcast: no switching, everything is transmitted to every node.
- Circuit switching: This came from the historical precedence that most early networks were telecommunication networks

Network Principles: Switching Schemes

- Packet Switching: The advent of digital technology and computers brought many new possibilities for telecommunication. “storage” and “processing” brought about innovations. For example, “Store and forward” network. Nothing new, digital version of Pony express.
- Frame relay: Frames of packets are relayed quickly through the network. Was a fancy a while ago. ATM is an example: Asynchronous Transmission Mode

Network Principles: Protocols

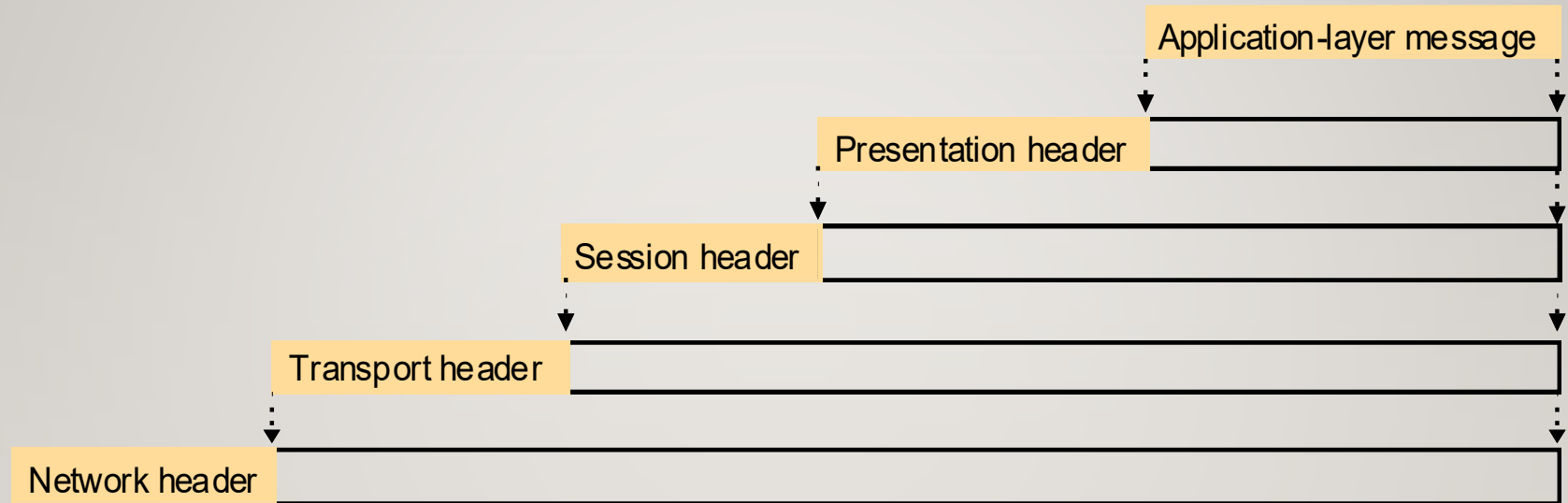
- What is a protocol? Is a well-known set of rules and formats to be used for communication between processes in order to perform a given task.
- A network protocol has two essential parts:
 - A specification of the sequence of messages that must be exchanged.
 - A specification of the format of the data in the messages.
- A protocol is implemented by a matched pair of software modules located in the sending and receiving computers.

Internet protocol :A little bit of history

- The Internet emerged from TWO decades of research and development work on wide area network in the USA. → ARPANET - first large scale network (~1970)
- Main contribution is the TCP/IP protocol suite
- TCP transmission control protocol
- IP Internet Protocol
- Many application level protocols emerged based on the TCP/IP foundation: Web (HTTP, HTTPS), email (SMTP, POP), file transfer (FTP, SFTP)

Figure 3.3

Encapsulation as it is applied in layered protocols



Protocol Layers

- Network software is arranged in a hierarchy of layers;
- Each layer presents an interface to the layer above it that extends the properties of the underlying communication system.
- Each layer is represented by a module in every computer connected to the network.
- Each layer of the connected computers seems to be interacting with the corresponding layer on the connected computer.
- Each layer provides a service to the layer above.
- At the bottom is the physical layer: communication medium: satellite comm channels, fiber, copper wire etc.

Protocol Suites

- A complete set of protocol layers is referred to as a protocol suite or a protocol stack, reflecting the layered architecture.
- ISO International standards organization's Open System Interconnection standard (OSI)
- Protocol layering brings substantial benefits in simplifying and generalization the software interfaces for access to the communication services of networks, but it also carries significant performance costs. Consider the costs...N layers, N transfers?

Figure 3.4 Protocol layers in the ISO Open Systems Interconnection (OSI) model

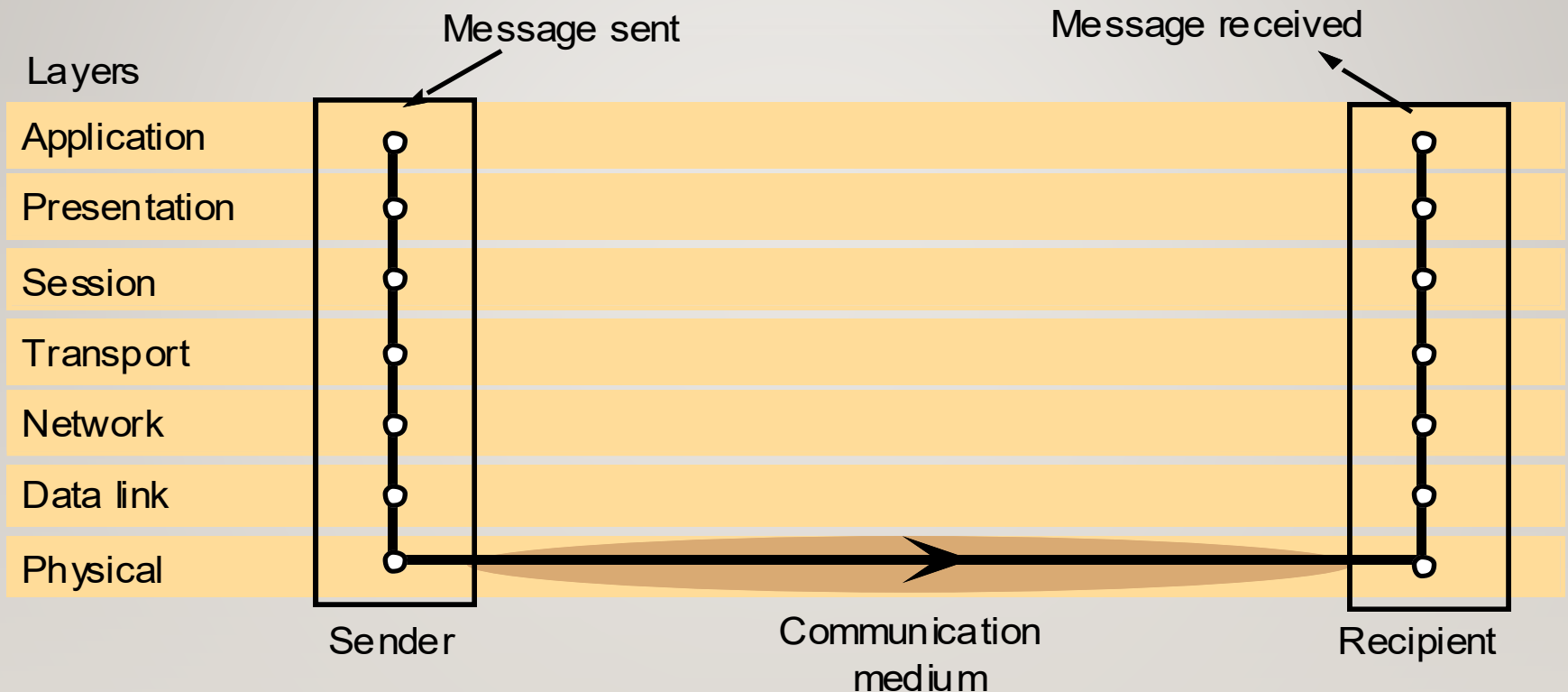


Figure 3.5

OSI protocol summary

Layer	Description	Examples
Application	Protocols that are designed to meet the communication requirements of specific applications, often defining the interface to a service.	HTTP, FTP , SMTP, CORBA IIOP
Presentation	Protocols at this level transmit data in a network representation that is independent of the representations used in individual computers, which may differ. Encryption is also performed in this layer, if required.	Secure Sockets (SSL),CORBA Data Rep.
Session	At this level reliability and adaptation are performed, such as detection of failures and automatic recovery.	
Transport	This is the lowest level at which messages (rather than packets) are handled. Messages are addressed to communication ports attached to processes, Protocols in this layer may be connection-oriented or connectionless.	TCP, UDP
Network	Transfers data packets between computers in a specific network. In a WAN or an internetwork this involves the generation of a route passing through routers. In a single LAN no routing is required.	IP, ATM virtual circuits
Data link	Responsible for transmission of packets between nodes that are directly connected by a physical link. In a WAN transmission is between pairs of routers or between routers and hosts. In a LAN it is between any pair of hosts.	Ethernet MAC, ATM cell transfer, PPP
Physical	The circuits and hardware that drive the network. It transmits sequences of binary data by analogue signalling, using amplitude or frequency modulation of electrical signals (on cable circuits), light signals (on fibre optic circuits) or other electromagnetic signals (on radio and microwave circuits).	Ethernet base- band signalling, ISDN

Packets, Ports and Addresses

- Packets have a header and data field
- Data is variable length, and header is fixed length (typically), limits such maximum transfer unit MTU can set the length of a packet.
- Ports: Provide network-independent message transport layer between a pair of network ports. Ports are software defined software defined destination points at a host computer.
- Example: Http: 80, ftp: 21, https 443, sftp (ssh) 22 are reserved
- Addresses are specified by network address of the host computer and the port number.
- Transport layer is responsible for delivering messages to destination with address as specified above.
- Next lets look at routing:

Routing

- Routing is a function that is required in all networks except LANs such as Ethernet that provides direct connection between all pairs of attached hosts.
- Delivery of packets is the collective responsibility of special dedicated computers called routers located at strategic points within a network.
- Best route for communication between two points is re-evaluated periodically taking into account current traffic and any faults such as broken connections and routers.
- A routing algorithm has two functions:
 1. Decide the route taken by each packet as it travels through the network.
 2. Dynamically update its knowledge of the network based on traffic monitoring and the detection of configuration changes or failures.
- A simple Routing Information Protocol (RIP): This is a basic algorithm that has been improved many folds and the information maintained at the routers also have increased. This has lead to smart and efficient routing.

Figure 3.7

Routing in a wide area network

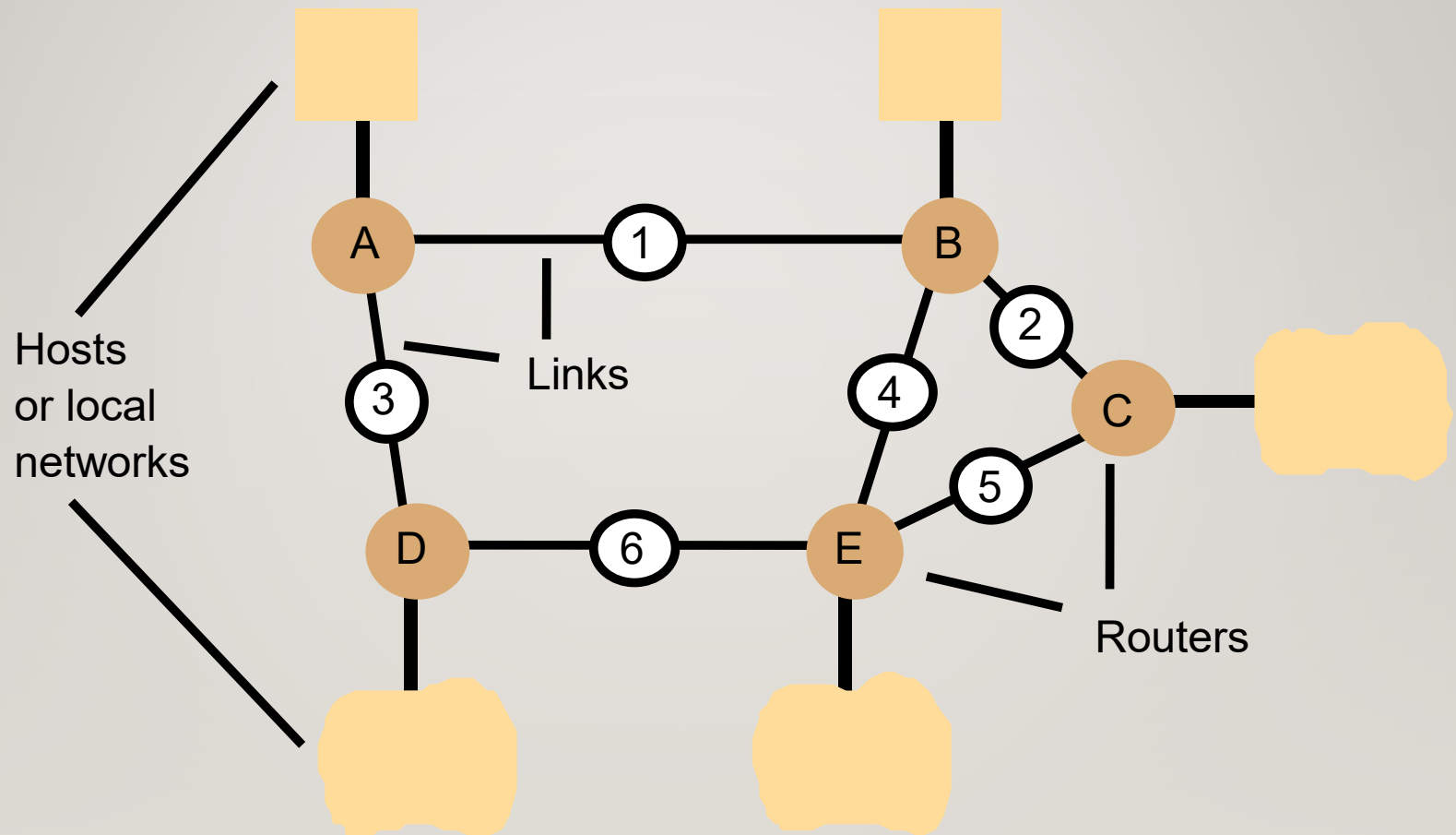


Figure 3.8

Routing tables for the network in Figure 3.7

Routings from A		
To	Link	Cost
A	local	0
B	1	1
C	1	2
D	3	1
E	1	2

Routings from B		
To	Link	Cost
A	1	1
B	local	0
C	2	1
D	1	2
E	4	1

Routings from C		
To	Link	Cost
A	2	2
B	2	1
C	local	0
D	5	2
E	5	1

Routings from D		
To	Link	Cost
A	3	1
B	3	2
C	6	2
D	local	0
E	6	1

Routings from E		
To	Link	Cost
A	4	2
B	4	1
C	5	1
D	6	1
E	local	0

Figure 3.9: Oldest distance-based algorithm

Pseudo-code for RIP routing algorithm

Send: Each t seconds or when Tl changes, send Tl on each non-faulty outgoing link.

Receive: Whenever a routing table Tr is received on link n :

```
for all rows  $Rr$  in  $Tr$  {  
  if ( $Rr.link \neq n$ ) {  
     $Rr.cost = Rr.cost + 1$ ;  
     $Rr.link = n$ ;  
    if ( $Rr.destination$  is not in  $Tl$ ) add  $Rr$  to  $Tl$ ;  
    // add new destination to  $Tl$   
  else for all rows  $Rl$  in  $Tl$  {  
    if ( $Rr.destination == Rl.destination$  and  
        ( $Rr.cost < Rl.cost$  or  $Rl.link == n$ ))  $Rl = Rr$ ;  
    //  $Rr.cost < Rl.cost$  : remote node has better route  
    //  $Rl.link = n$  : remote node is more authoritative  
  }  
}  
}
```

Figure 3.11

Tunnelling for IPv6 migration

IPv6 encapsulated in IPv4 packets

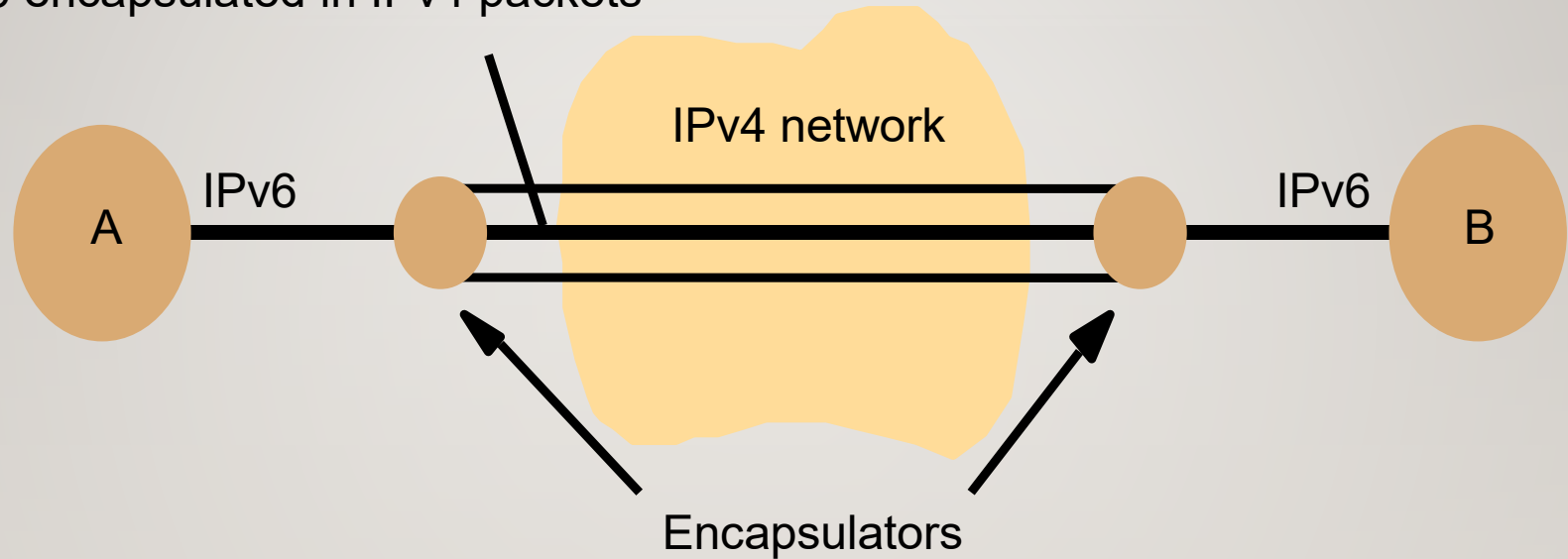


Figure 3.13 Encapsulation in a message transmitted via TCP over an Ethernet

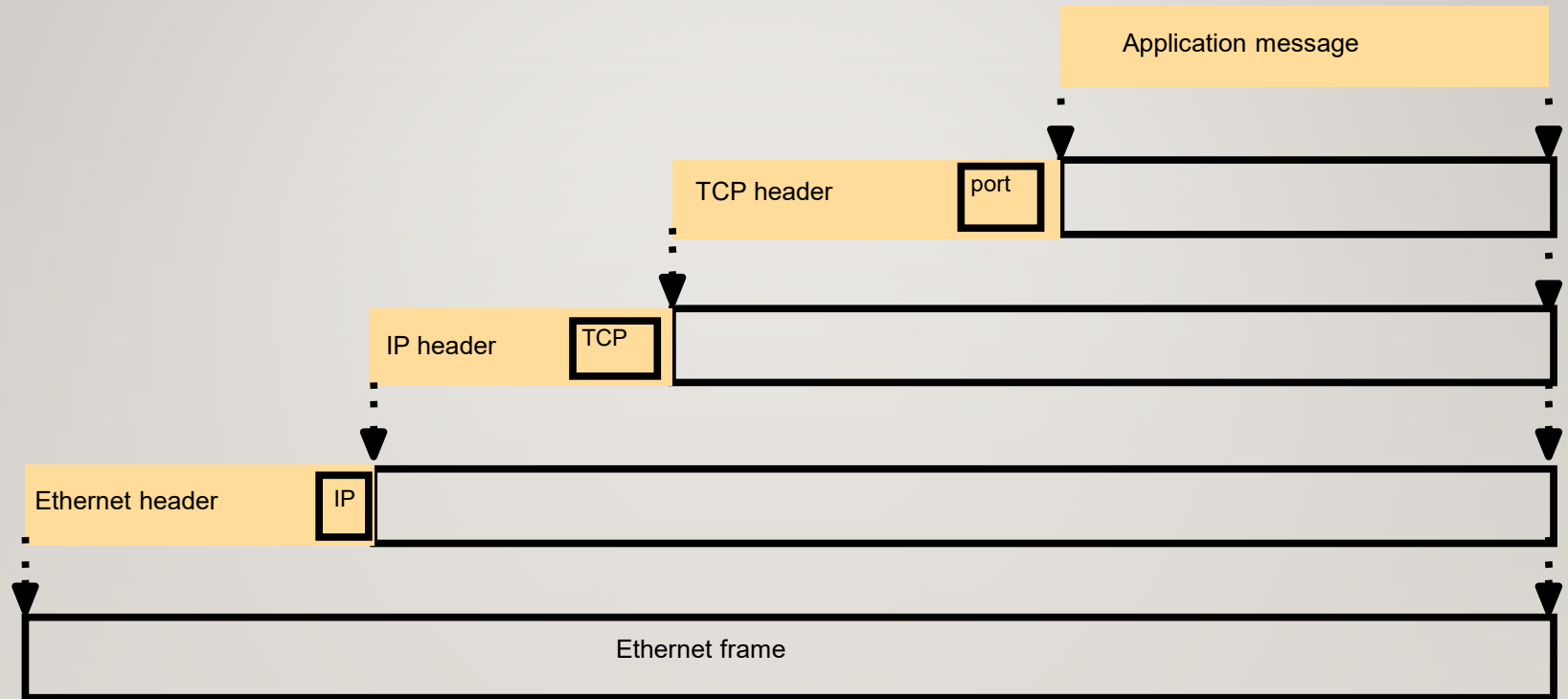


Figure 3.14 The programmer's conceptual view of a TCP/IP Internet

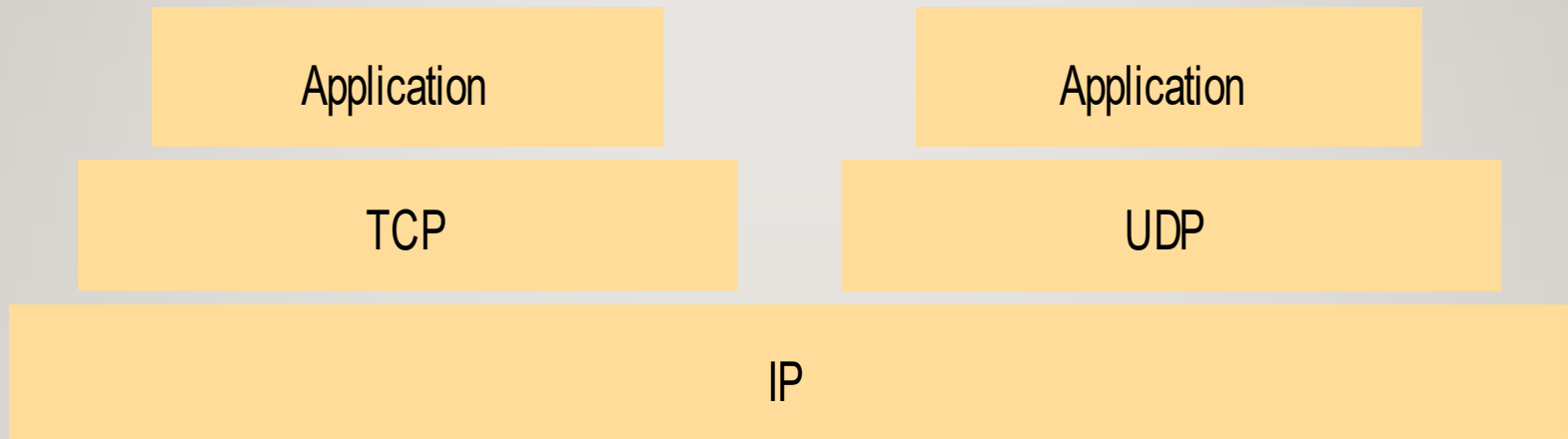


Figure 3.15

Internet address structure, showing field sizes in bits

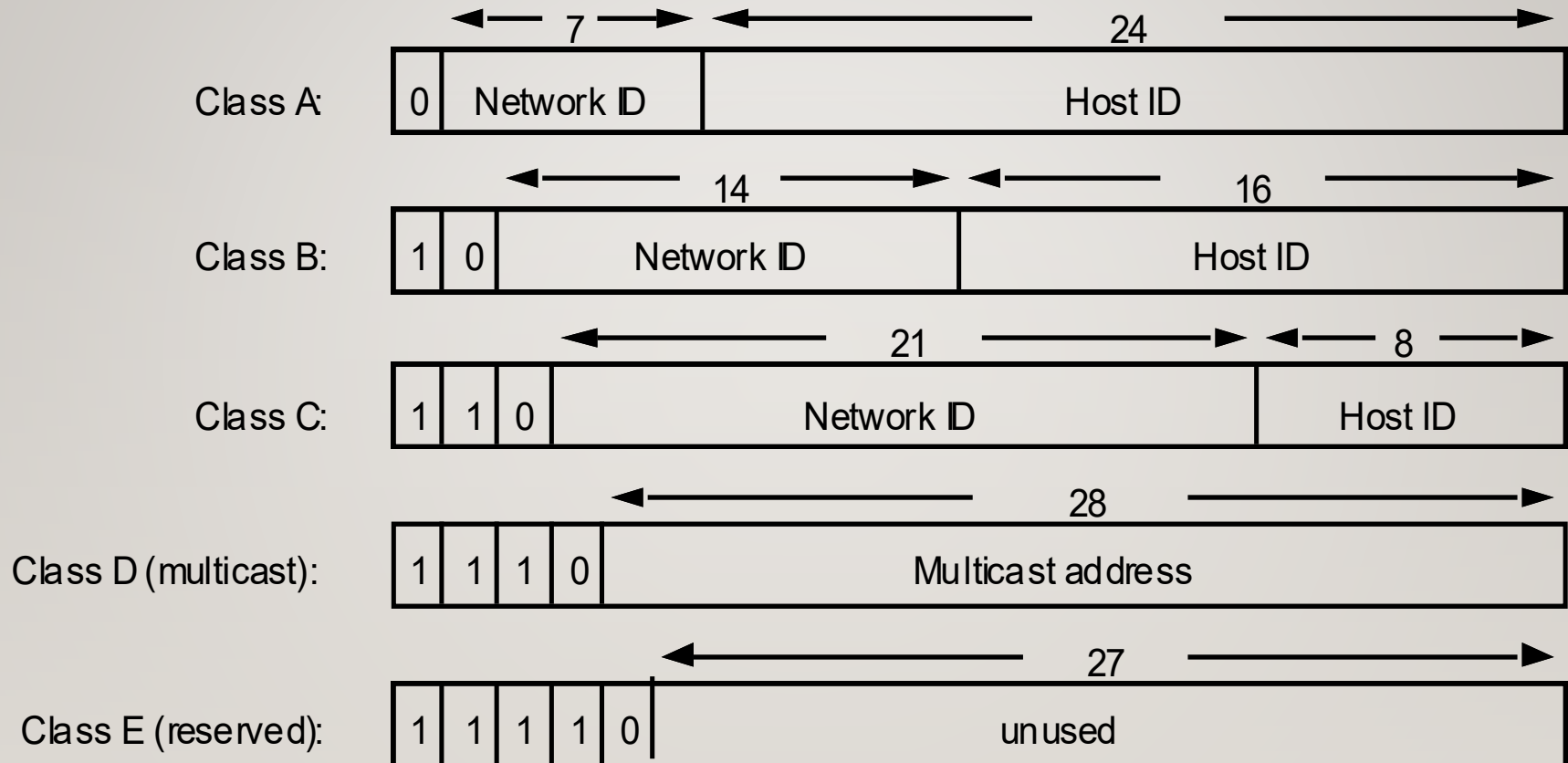


Figure 3.16

Decimal representation of Internet addresses

	octet 1	octet 2	octet 3		Range of addresses
	Network ID		Host ID		
Class A:	1 to 127	0 to 255	0 to 255	0 to 255	1.0.0.0 to 127.255.255.255
	Network ID		Host ID		
Class B:	128 to 191	0 to 255	0 to 255	0 to 255	128.0.0.0 to 191.255.255.255
		Network ID		Host ID	
Class C:	192 to 223	0 to 255	0 to 255	1 to 254	192.0.0.0 to 223.255.255.255
	Multicast address				
Class D (multicast):	224 to 239	0 to 255	0 to 255	1 to 254	224.0.0.0 to 239.255.255.255
Class E (reserved):	240 to 255	0 to 255	0 to 255	1 to 254	240.0.0.0 to 255.255.255.255

Figure 3.17

IP packet layout

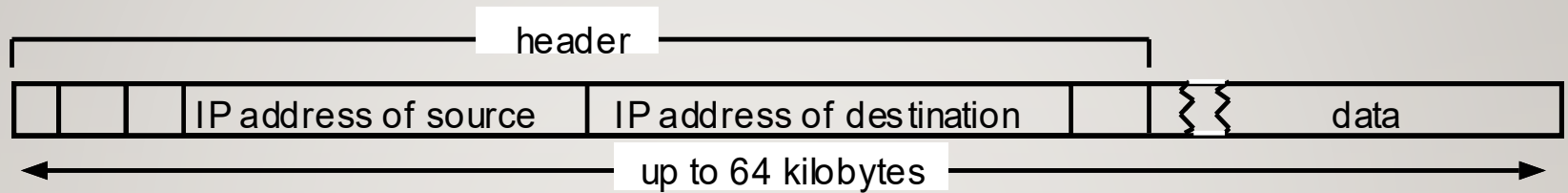


Figure 3.18

A typical NAT-based home network

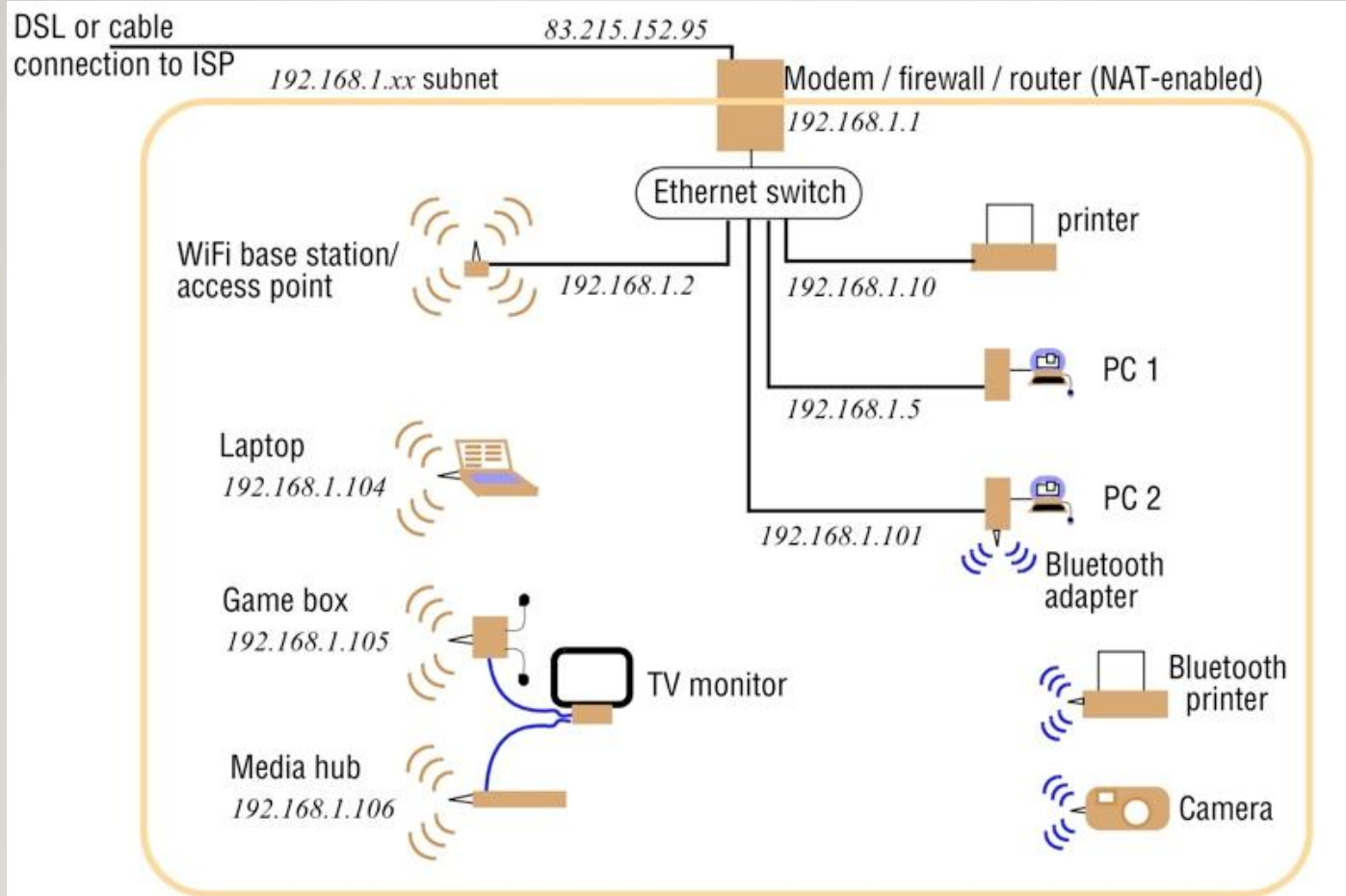


Figure 3.19

IPv6 header layout

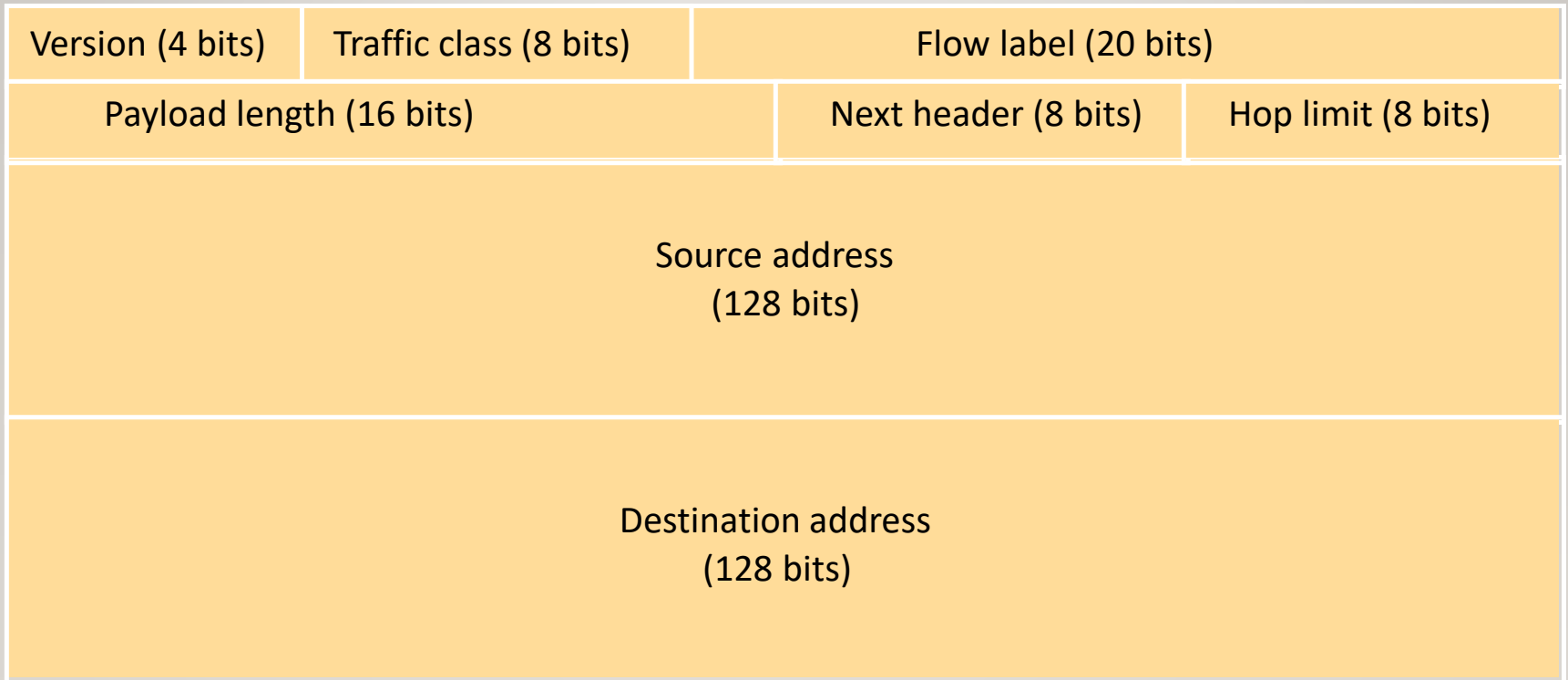


Figure 3.22

IEEE 802 network standards

<i>IEEE No.</i>	<i>Name</i>	<i>Title</i>	<i>Reference</i>
802.3	Ethernet	CSMA/CD Networks (Ethernet)	[IEEE 1985a]
802.4		Token Bus Networks	[IEEE 1985b]
802.5		Token Ring Networks	[IEEE 1985c]
802.6		Metropolitan Area Networks	[IEEE 1994]
802.11	WiFi	Wireless Local Area Networks	[IEEE 1999]
802.15.1	Bluetooth	Wireless Personal Area Networks	[IEEE 2002]
802.15.4	ZigBee	Wireless Sensor Networks	[IEEE 2003]
802.16	WiMAX	Wireless Metropolitan Area Networks	[IEEE 2004a]

Figure 3.23

Ethernet ranges and speeds

	<i>10Base5</i>	<i>10BaseT</i>	<i>100BaseT</i>	<i>1000BaseT</i>
Data rate	10 Mbps	10 Mbps	100 Mbps	1000 Mbps
<i>Max. segment lengths:</i>				
Twisted wire (UTP)	100 m	100 m	100 m	25 m
Coaxial cable (STP)	500 m	500 m	500 m	25 m
Multi-mode fibre	2000 m	2000 m	500 m	500 m
Mono-mode fibre	25000 m	25000 m	20000 m	2000 m

Figure 3.24

Wireless LAN configuration

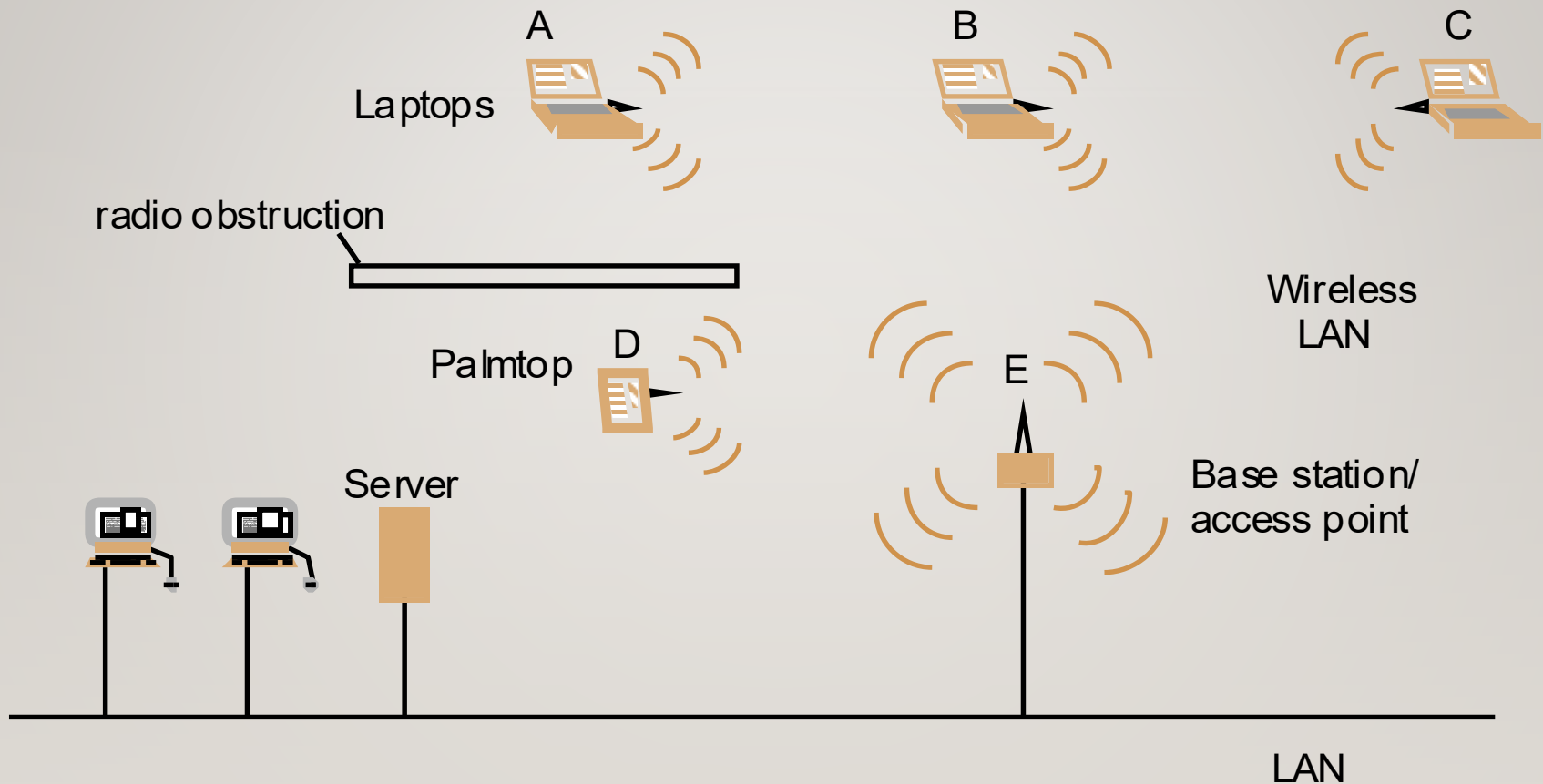
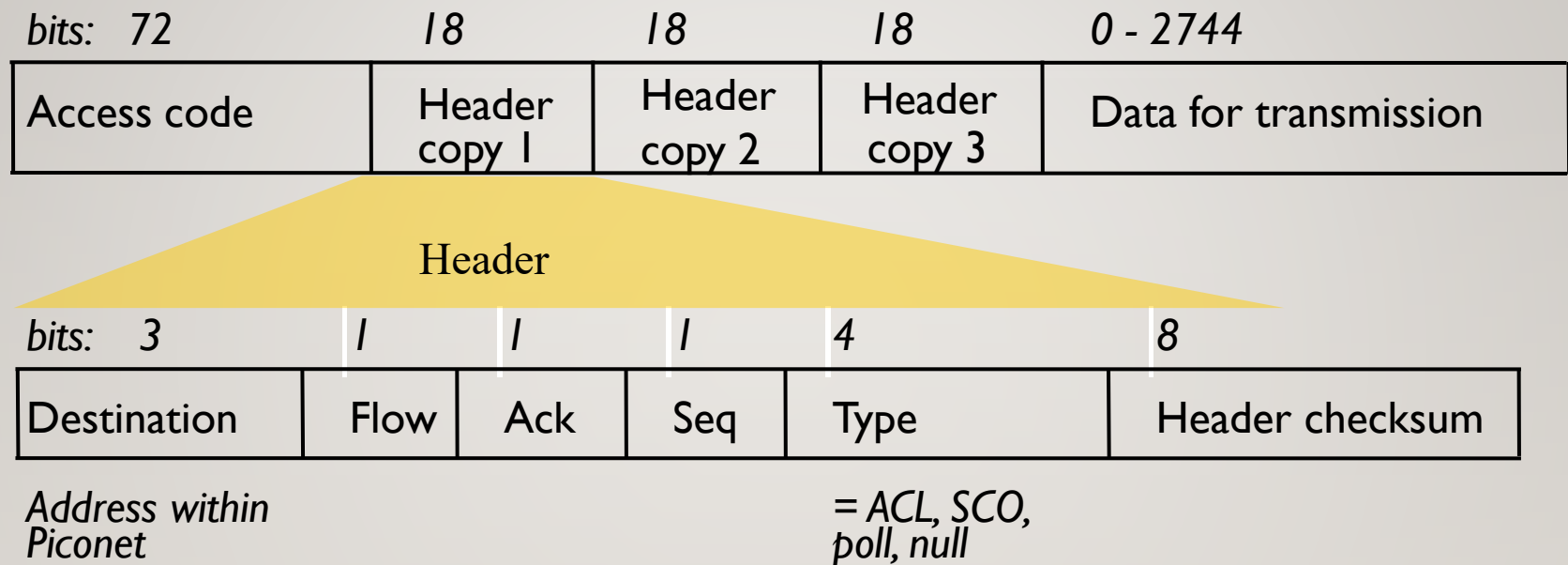


Figure 3.25

Bluetooth frame structure



SCO packets (e.g. for voice data) have a 240-bit payload containing 80 bits of data triplicated, filling exactly one timeslot.